Following the Flow: PFAS Detection Patterns Across Colorado

Maisie Moline

Colorado State University

Watershed Problem Analysis (WR 440)

May 13, 2026

Following the Flow: PFAS Detection Patterns Across Colorado

Per- and polyfluoroalkyl substances (PFAS) are a large class of synthetic chemicals that have been manufactured since the 1940s for a multitude of purposes (Glüge et al., 2020). Their widespread use and environmental persistence have led to growing concern about their presence in natural and drinking water systems. Many PFAS compounds contain strong carbon–fluorine bonds that make them highly resistant to chemical, biological, and environmental degradation (Abunada et al., 2020). These characteristics result in PFAS persistence, mobility, and toxicity in the environment.

Since the 1940s, PFAS have become commonplace across many industrial and consumer sectors due to various useful properties. Some of these properties include: non-flammable, non-reactive, very stable, low surface tension, and hydrophobic. Notable Industry branches utilizing PFAS include aerospace, mining, energy, oil and gas, and electronics. Some common uses include electronics, textiles, automotive, cleaning compositions, fire-fighting foam, laboratory supplies, personal care products, pesticides, plastic and rubber, sports articles, filter membranes, and pharmaceuticals, to name a few (Glüge et al., 2020). The numerous uses coupled with its useful properties make it hard to eliminate PFAS use. New kinds of PFAS are still being developed, as others are being phased out. Few scientific studies on human and environmental health support the new PFAS.

PFAS exposure has negative impacts on human health. This exposure occurs through a variety of pathways, such as living near sites with contaminated soil or water. This pollution occurs through runoff or leaching. Contaminated water causes exposure through its use for drinking water and agriculture. Additionally, inhaling PFAS— such as fire-fighting foam, manufacturing chemicals, or contaminated dust— can result in negative health impacts (Centers for Disease Control and Prevention (CDC), 2025). Research is ongoing regarding the health effects of PFAS. There is evidence of increasing PFAS being associated with increases in cholesterol, lower antibody response, changes in liver enzymes, decreases in birth weight, and increased risk of certain types of cancer. Health impacts depend on exposure factors, physical characteristics, and other socioeconomic determinants (Barton et al., 2025 & CDC, 2025).

As research has increasingly documented the harmful effects of PFAS, regulatory agencies have begun establishing testing protocols and setting limits for these contaminants. Many of these regulations have been established, but are not yet in effect. In the United States, the Final PFAS National Primary Drinking Water Regulation was published in 2024, outlining regulations for six PFAS that public water systems must monitor for. Under this regulation, initial monitoring must be done by 2027, followed by ongoing maintenance and public information. By 2029, public water systems must move to implement solutions reducing PFAS if any violations are detected. The EPA's motivations behind this ruling are to prevent PFAS exposure in drinking water for around 100 million people, preventing deaths and illnesses (U.S. Environmental Protection Agency, n.d.).

While the 2024 PFAS National Primary Drinking Water Regulation establishes enforceable limits for six PFAS, the EPA also relies on broader monitoring programs to identify additional contaminants of concern. One of the most important of these efforts is the Unregulated Contaminant Monitoring Rule (UCMR), which requires public water systems to track emerging contaminants such as PFAS. UCMR 5 specifically requires monitoring by PWSs serving more than 3,300 people for 29 per- and polyfluoroalkyl substances. This monitoring timeframe spanned from 2023 to 2025, with results posted to a national dataset (U.S. Environmental Protection Agency, 2021). This data set will be used to answer the question of how PFAS detections in public drinking water systems are distributed across Colorado, and how detection levels differ by system size and water source. To investigate this question, this study will compare PFAS detection levels among public water systems serving different population sizes and using different source water types. Through this analysis, potential gaps in monitoring coverage may also be identified, highlighting limitations in the dataset.

Site Description

Colorado is a state characterized by its sunny, semiarid climate and outdoor features. Colorado's topography can generally be divided into three major regions: the plains in the East, the mountains in the middle of the state, and the plateaus and mesas in the West. Notable Landcover in Colorado includes 24.5 million acres of forests and 30 million acres of farmland. The average elevation in Colorado is 6,800 feet. In the Rocky Mountains, elevations begin at 7,000 feet and climb to 9,000-14,000 feet, including some of the tallest peaks in the region. Across Colorado, average annual precipitation is about 17 inches, with the most precipitation falling in the mountains (Colorado Water Center, n.d.). Temperatures differ across the state due to differences in elevation. In the summer, areas in the lower elevation areas reach 95 degrees Farenheight, while only reaching 70 ℉ in the peaks of the mountains. In the winter, temperatures can dip to 5 ℉ in the mountains while lower elevations are closer to 20 ℉ (WeatherSpark, n.d.).

Colorado is considered a headwater state because four major rivers originate in the Colorado Rocky Mountains: the Colorado, Rio Grande, Arkansas, and Platte (CWCB, 2023 & Figure 2). The Colorado River Basin serves the most people among the four basins, reported as providing drinking water for around 40 million people (Mullane, 2023), but the South Platte serves the most people in Colorado. Across the basins, there is variability in their topography, precipitation, water uses, and more (CWCB, 2023 & Legislative Council Staff, 2024). The mismatch between water availability and water demand makes it essential to maintain water quality standards to support ecosystem and human health in these basins. Water's economic impact is another important reason for water monitoring efforts. Just in Colorado, as published by the Colorado Water Conservation Board (CWCB), the state's economy is driven by agriculture, worth $47 billion, and water-based recreation, contributing around $19 billion. By sector, Agriculture uses the most water (90%), followed by municipalities (7%), then large industry (3%). In these sectors, 17% is sourced from groundwater, and 83% is sourced from surface water (2023).

While most of the population is located East of the Continental Divide, 80% of the water originates in rivers West of the Divide and is transferred through several tunnel and ditch diversions that move it to the Eastern Slope. The majority of Public Water Systems are to the East of the Divide. Despite the continued population growth, overall water demand has remained steady due to water conservation and improved infrastructure. (CWCB, 2023). Within Colorado, there are approximately 2,051 water providers that differ in size, capacity, and responsibilities, with some systems serving as few as 25 people. An estimated 227 cities and towns in Colorado manage a public water system. While these public water systems create their own plans and requirements, they must comply with federal and state drinking water regulations as well as with statutes and compacts governing water use in the West. Regulations established under the Safe Drinking Water Act require routine monitoring for regulated contaminants and periodic monitoring for emerging contaminants that are not yet fully regulated (Colorado Department of Local Affairs, Division of Local Government, n.d.). The Safe Drinking Water Act (SDWA) requires the EPA, once every five years, to issue a list of unregulated contaminants that PWSs should monitor, also known as the Unregulated Contaminant Monitoring Rule (UCMR). UCMR

5 specifically includes monitoring for multiple PFAS compounds in public drinking water systems, which improves understanding of the prevalence and quantity of 29 per- and polyfluoroalkyl substances (PFAS) (U.S. Environmental Protection Agency, 2021). The PWSs included in UCMR 5 are mapped in Figure 1, Site Map.

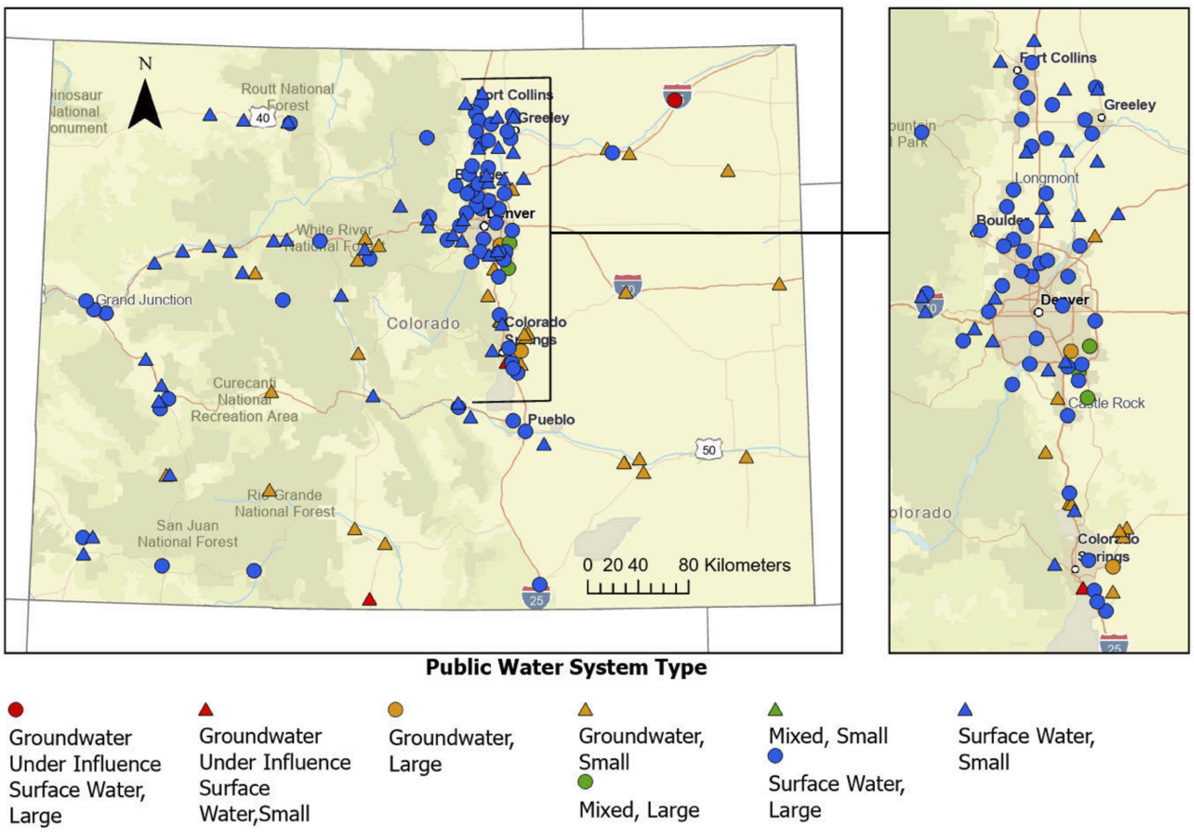

Figure 1. Study Area in Colorado showing public water systems included in this study. Points represent public water systems of sampled under EPA UCMR 5 colored by source water type and symbolized by water system size (Large>10,000 people served; Small<=10,000 people served). An inset map highlights the Front Range Urban Corridor where water systems are densely concentrated.

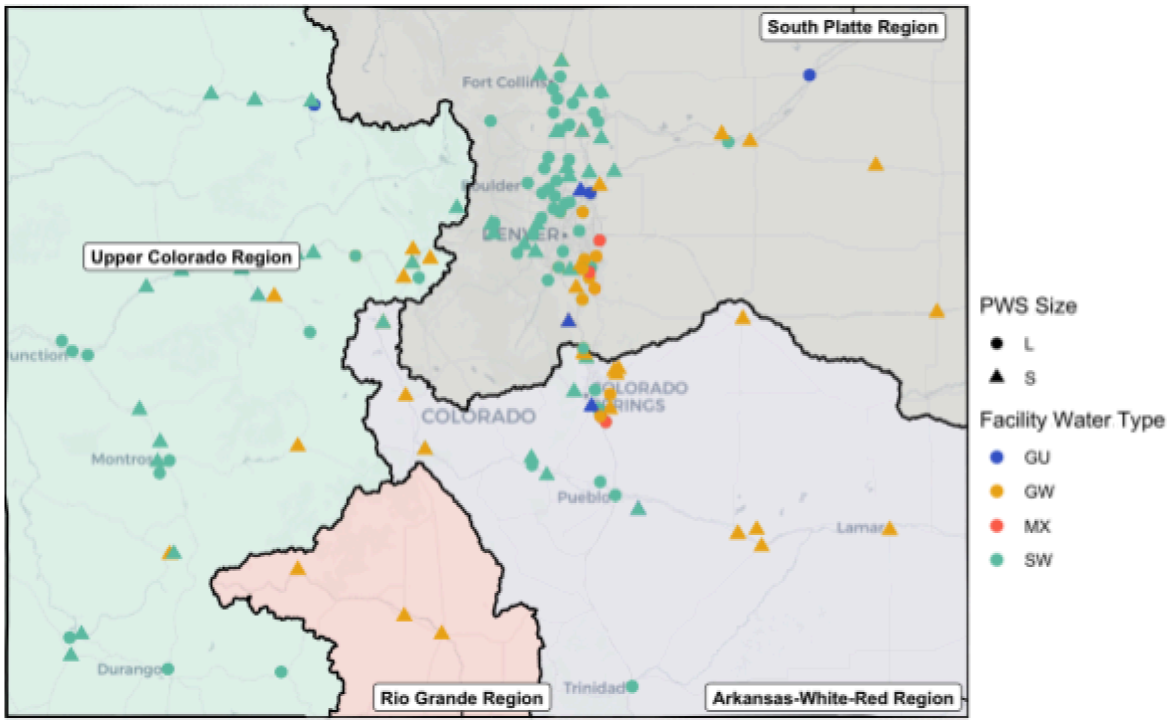

Figure 2. Study Area in Colorado showing public water systems in their respective watersheds. The map shows the four major watersheds in Colorado, including the Upper Colorado, Rio Grande, South Platte, and Arkansas. Points show public water systems (PWS) sampled under UCMR 5, with facility water types colored dark blue for groundwater influence, yellow for groundwater, teal for surface water, and red for mixed.

Methods

Data were obtained from the EPA Fifth Unregulated Contaminant Monitoring Rule Data Finder filtered to the state of Colorado (U.S. Environmental Protection Agency [EPA], 2026). This data was entered over a period of 3 years, spanning anywhere from 2023 to 2025. The PFAS are trimmed to 6 contaminants: PFBA, PFBS, PFHxA, PFHxS, PFOA, PFOS. This selection is based on the quantity of data available and PFAS with established limits regulated by the EPA. Three main objectives were selected to investigate how PFAS detections in public drinking water systems are distributed across Colorado, and how detection levels differ by system size and water source. These objectives include summarizing the location and concentration of PFAS detections in public drinking water systems across Colorado, comparing PFAS detection levels among public water systems of different sizes, and comparing PFAS detections between groundwater-supplied and surface water-supplied drinking water facilities. Portions of the R code development and debugging process were assisted by AI-based tools (Microsoft Copilot), which were used to support code refinement and troubleshooting. All final analyses, interpretations, and results were reviewed and verified by the author.

Objective 1:

To summarise the location and concentration of PFAS detections, the EPA dataset is joined with public water system (PWS) metadata using the PWS ID as the key. This metadata should include the latitude and the longitude of the public water systems. Before beginning this analysis, the data are filtered to PFAS detections, mutating “<MRL”(Minimum Risk Level) to 0 values. Using the ggplot2 package (V 4.0.0; Wickham, 2016), ggspatial package (V 1.1.10; Dunnington, 2025), and sf package (V 1.1.0; Pebesma & Bivand, 2023) in R (V 4.41; R Core Team, 2024), detections are mapped in Colorado, and points are colored by facility water type and shaped by PWS size. The top five systems with the highest number of detections are labeled on the map for each PFAS type.

Additional information about these PWS is summarised in a table, including the number of reports, the number of detections <MRL, and detection frequency. Detection frequency is calculated by dividing the number of detections by the number of reports grouped by the water system.

Objective 2:

Objective 2: To determine whether PFAS levels differ across public water systems of different sizes, the column “PWS.Size” is used from the EPA dataset. This category classifies the

PWS as either small, with fewer than 3000 consumers, or larger, with more than 3,000 consumers. Before beginning calculations, detections considered “<MRL” are replaced with 0 values for this analysis. Due to the non-normal distribution of the data, the Wilcoxon rank-sum test is used to compare PFAS distributions between large and small PWS systems and determines if significant at an alpha level of 0.05. These findings are summarised in a table with the dyplr package (V 1.1.4; Wickham et al., 2023) and supported visually.

Objective 3:

Objective 3: To determine whether PFAS levels differ across facility water type —surface, ground, mixed, and under groundwater influence— the column Facility.Water.Type is used. Before beginning calculations, detections considered “<MRL” are replaced with 0 values for this analysis. Differences in PFAS concentrations across facility water types are evaluated using a Kruskal–Wallis test because the data are non-normal. When significant at an alpha level of 0.05, pairwise comparisons are conducted using Wilcoxon rank-sum tests with Benjamini–Hochberg adjustment for multiple comparisons. Benjamin-Hochberg controls for false discovery rate, reducing the probability of making Type I errors (false positives) among the rejected null hypotheses, especially in studies with many tests. Results from these tests are summarised in a table and supported visually.

Results

Objective 1

The first objective was to summarize the location and concentration of PFAS detections in public drinking water systems across Colorado. The public water systems with the most frequent detections were Todd Creek Village, Brighton, and Thornton (Table 2). For each PFAS compound (Figures 3-8), all detections were mapped across Colorado. The five public water systems with the highest number of detections for that compound were identified and labeled on the map to highlight systems with the most frequent detection occurrence. All six PFAS were identified as part of the top five at the following PWSs: Brighton, Thornton, and Widefield. Four PFAS types were detected at Parker (PFBA, PFBS, PFHxA, PFOA) and Todd Creek Village (PFBA, PFBS, PFHxS, PFOS). Although not always in the top five detections labeled on the maps, Snake River WD detected all but one (PFBA) PFAS types (Table 1).

Table 1. Displays if PFAS type was detected (TRUE ✔) or absent (FALSE ✘) at the public water system (PWS). The subset was picked based on systems most frequently observed on the mapped figures to summarise findings from the mapped figures.

PWS	PFBA	PFBS	PFHxA	PFHxS	PFOA	PFOS
Brighton	✔	✔	✔	✔	✔	✔
Thornton	✔	✔	✔	✔	✔	✔
Widefield	✔	✔	✔	✔	✔	✔
Parker	✔	✔	✔	✘	✔	✘
Snake River	✘	✔	✔	✔	✔	✔
Todd Creek Village	✔	✔	✘	✔	✘	✔

Table 2. Summarises PFAS occurrence at different PWS. In this table are the number of samples for the PWS, the number of detections greater than the minimum reporting level, and the percent of total samples that were greater than the MRL. This PWS subset builds off sites selected for Table 1 and arranges them in descending order for percent above the minimum reporting level (MRL).

PWS	Samples	Number Detects	Percent Above MRL (%)
TODD CREEK VILLAGE	24	18	75
BRIGHTON	90	59	66
THORNTON	42	26	62

SNAKE RIVER	24	9	38
WIDEFIELD	144	45	31
PARKER	156	20	13

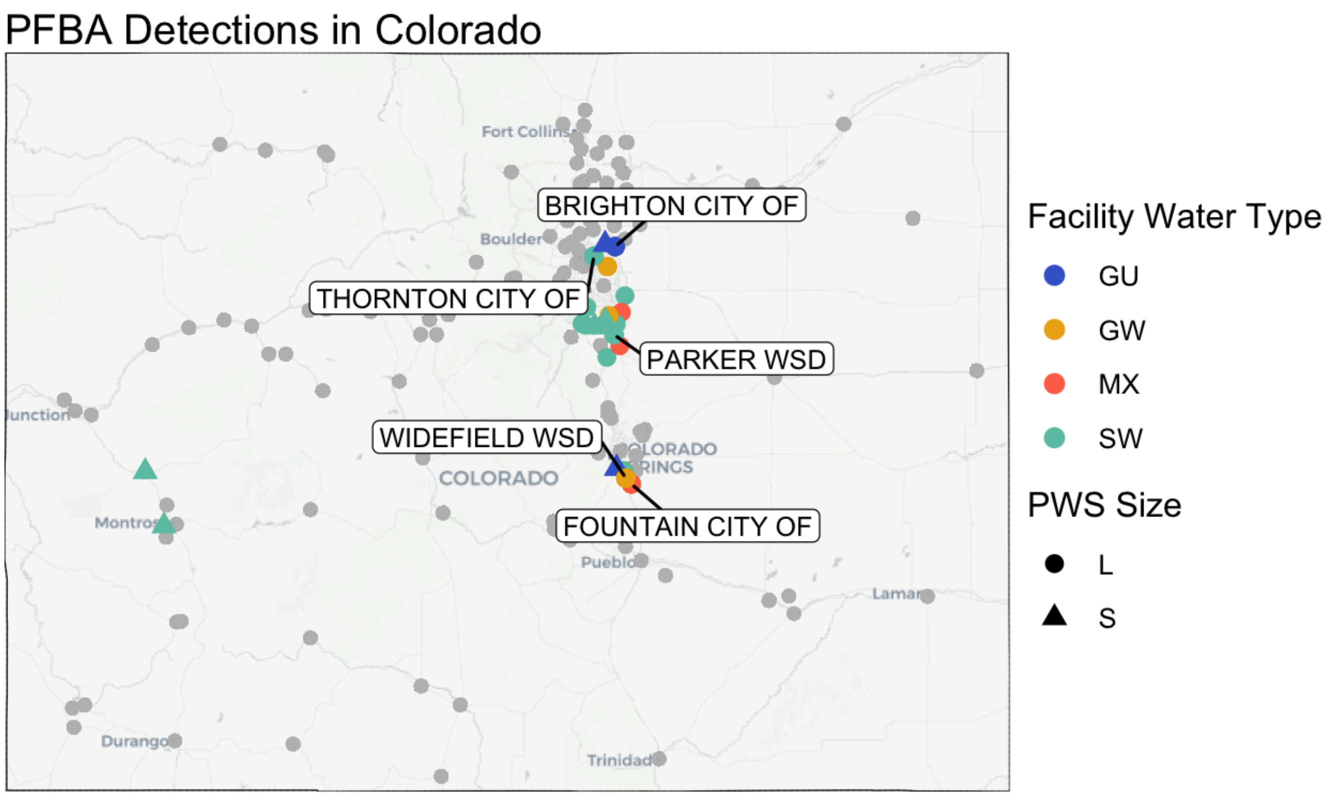

Figure 3. Spatial distribution of PFBA detections in Colorado public water systems. Points are colored by facility water type (groundwater, surface water, groundwater under the influence, or mixed) and shaped by system size (small or large). Grey points represent no detection. The five public water systems with the highest number of detections for that compound were identified and labeled on the map to highlight systems with the most frequent occurrence. Facility water types are classified as groundwater (GW), groundwater under the influence of surface water (GU), surface water (SW), and mixed sources (MX).

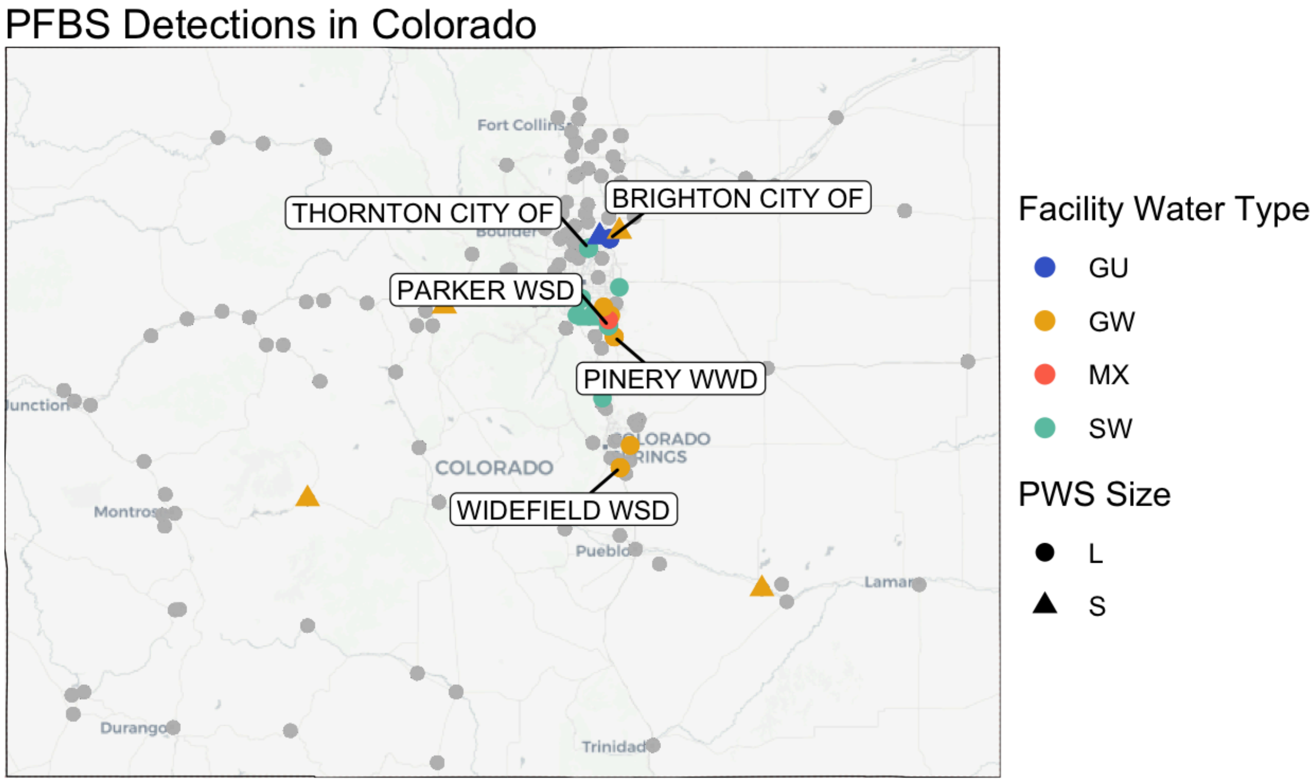

Figure 4. Spatial distribution of PFBS detections in Colorado public water systems. Points are colored by facility water type (groundwater, surface water, groundwater under the influence, or mixed) and shaped by system size (small or large). Grey points represent no detection. The five public water systems with the highest number of detections for that compound were identified and labeled on the map to highlight systems with the most frequent occurrence. Facility water types are classified as groundwater (GW), groundwater under the influence of surface water (GU), surface water (SW), and mixed sources (MX).

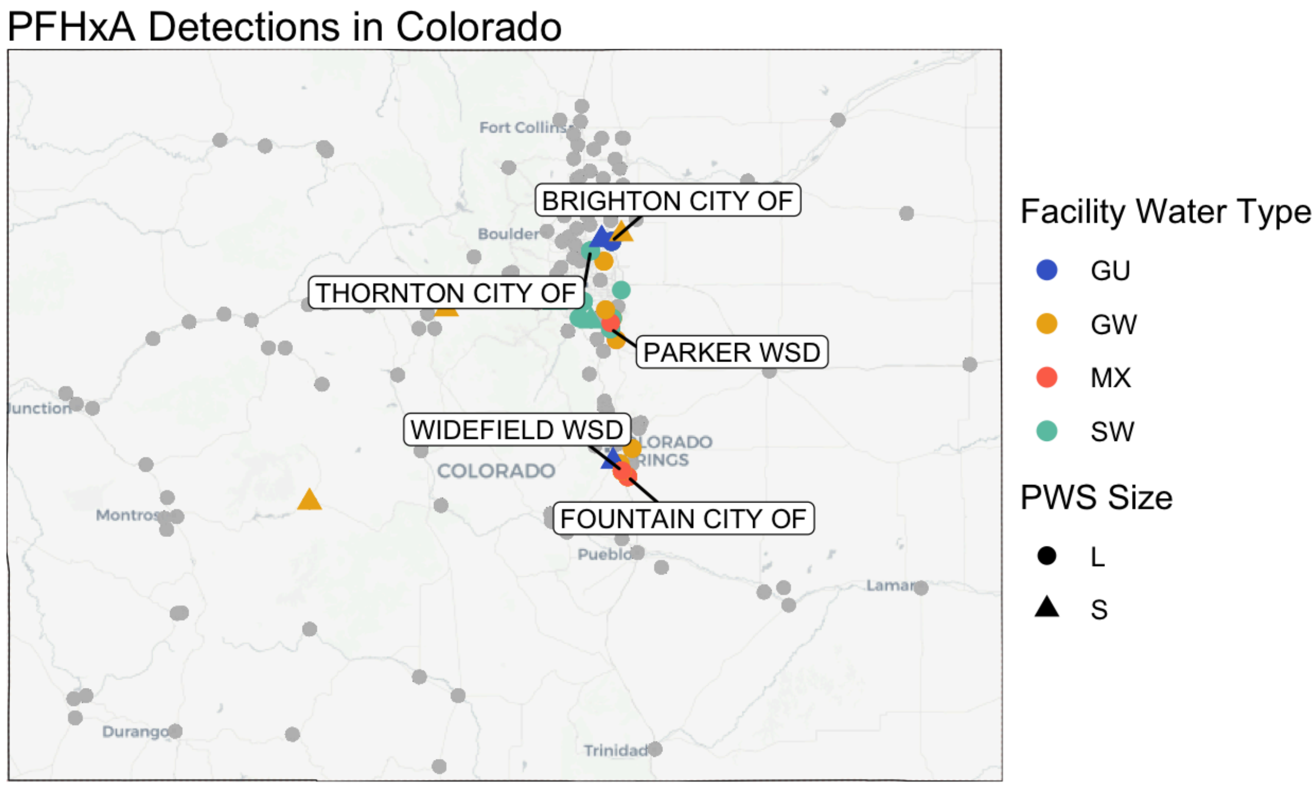

Figure 5. Spatial distribution of PFHxA detections in Colorado public water systems. Points are colored by facility water type (groundwater, surface water, groundwater under the influence, or mixed) and shaped by system size (small or large). Grey points represent no detection. The five public water systems with the highest number of detections for that compound were identified and labeled on the map to highlight systems with the most frequent occurrence. Facility water types are classified as groundwater (GW), groundwater under the influence of surface water (GU), surface water (SW), and mixed sources (MX).

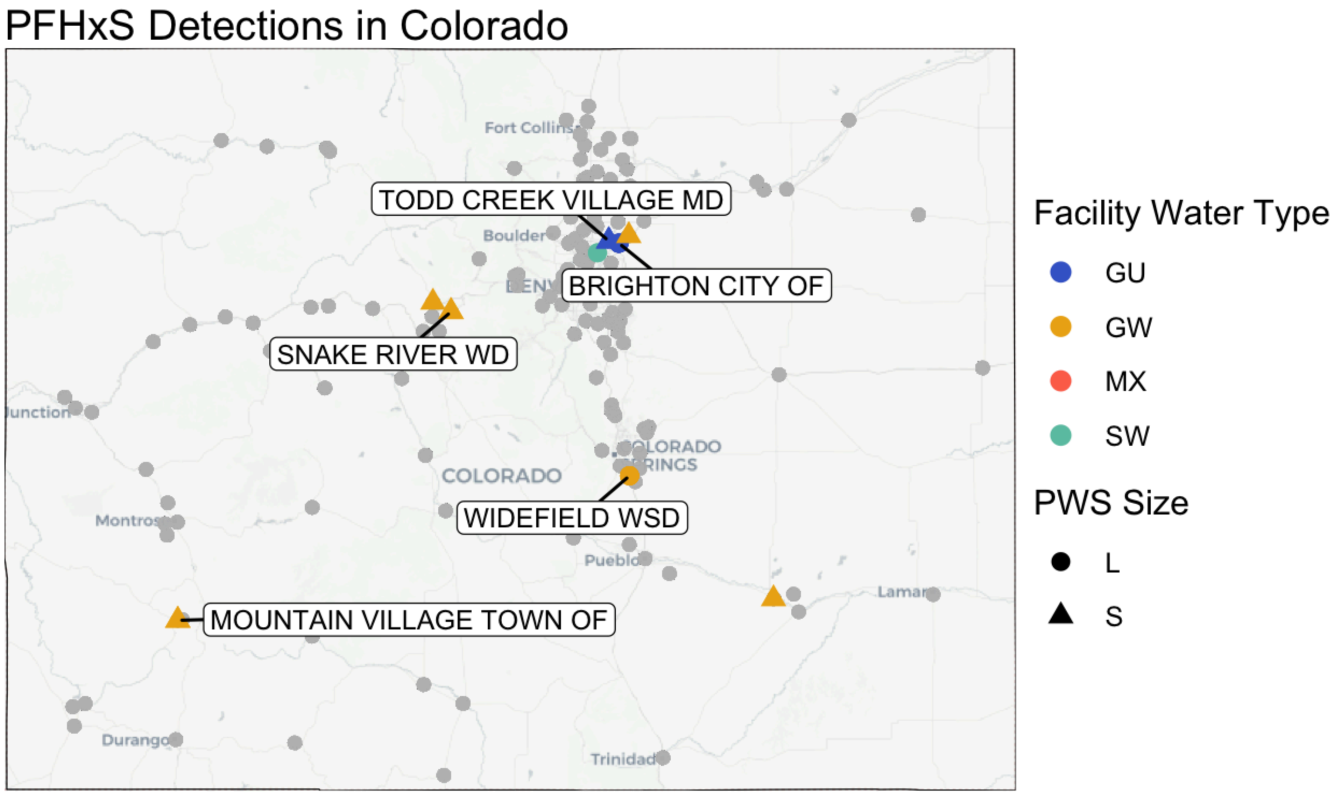

Figure 6. Spatial distribution of PFHxS detections in Colorado public water systems. Points are colored by facility water type (groundwater, surface water, groundwater under the influence, or mixed) and shaped by system size (small or large). Grey points represent no detection. The five public water systems with the highest number of detections for that compound were identified and labeled on the map to highlight systems with the most frequent occurrence. Facility water types are classified as groundwater (GW), groundwater under the influence of surface water (GU), surface water (SW), and mixed sources (MX).

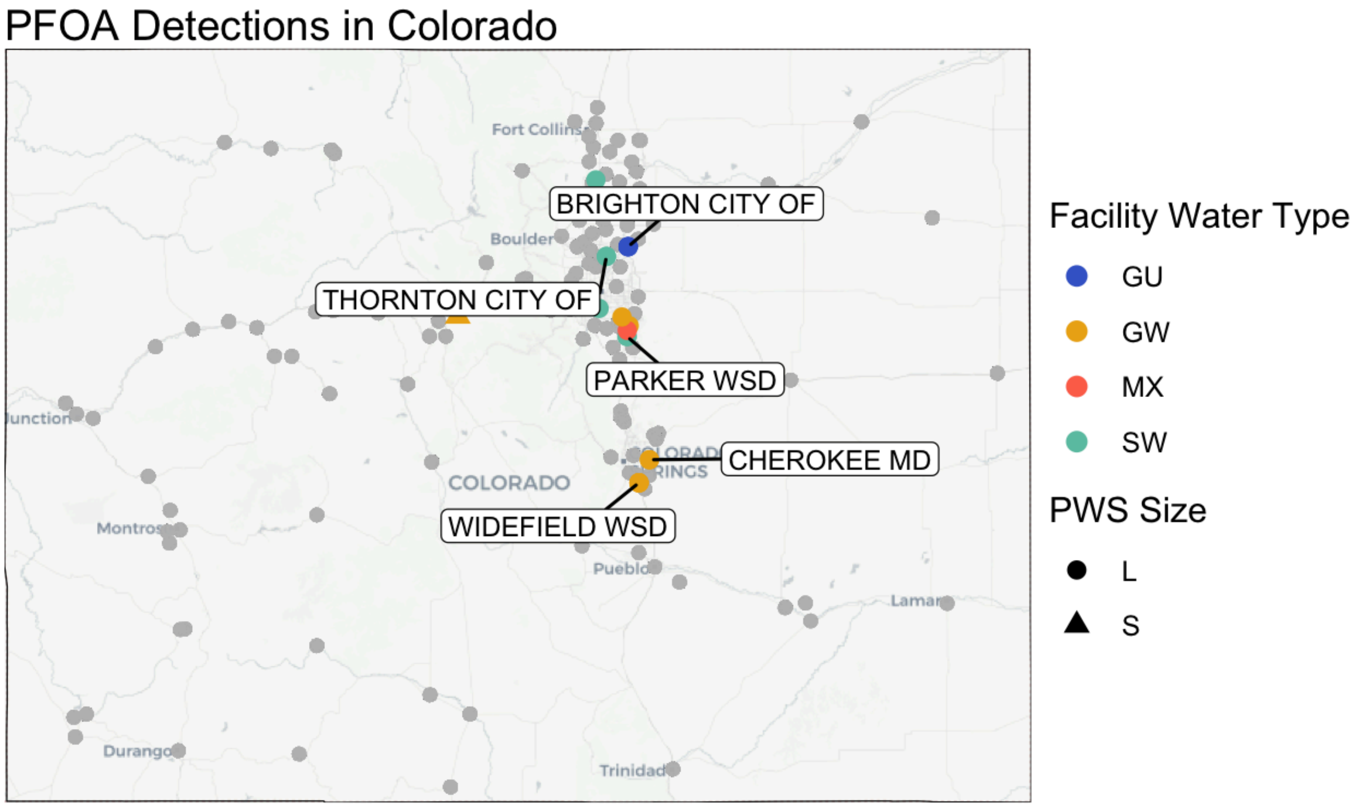

Figure 7. Spatial distribution of PFOA detections in Colorado public water systems. Points are colored by facility water type (groundwater, surface water, groundwater under the influence, or mixed) and shaped by system size (small or large). Grey points represent no detection. The five public water systems with the highest number of detections for that compound were identified and labeled on the map to highlight systems with the most frequent occurrence. Facility water types are classified as groundwater (GW), groundwater under the influence of surface water (GU), surface water (SW), and mixed sources (MX).

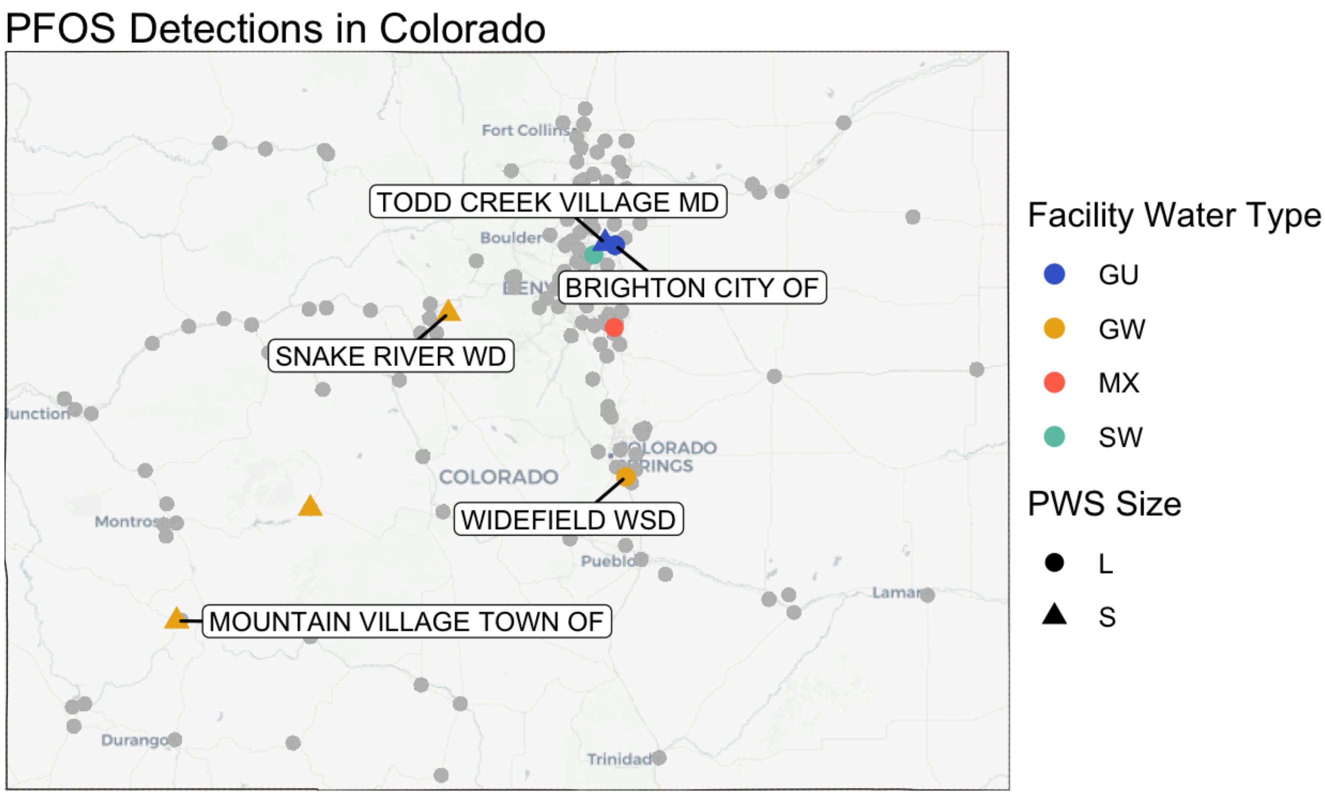

Figure 8. Spatial distribution of PFOS detections in Colorado public water systems. Points are colored by facility water type (groundwater, surface water, groundwater under the influence, or mixed) and shaped by system size (small or large). Grey points represent no detection. The five public water systems with the highest number of detections for that compound were identified and labeled on the map to highlight systems with the most frequent occurrence. Facility water types are classified as groundwater (GW), groundwater under the influence of surface water (GU), surface water (SW), and mixed sources (MX).

Objective 2

The second objective was to compare PFAS detection levels among public water systems of different sizes. Table 3 displays results from the Kruskal-Wallis rank sum test, comparing whether medians differ across PWS systems of different sizes. Results indicated PFBA, PFBS, PFHxA, PFOA, and the overall test, including all PFAS, are significant ($p<0.05$). These differences are highlighted visually in Figure 9 and 10. PFHxS and PFOS did not differ significantly across different sizes of water systems. Figure 9 shows that PFOA only has detections over the Minimum Reporting Level for large systems. The max contaminant level for PFBA, PFHxS, PFOA, and PFHxA is higher for large systems, while the max contaminant level for PFBS and PFOS is higher for small systems.

Table 3. Kruskal-Wallis Rank Sum Test for different-sized PWS systems. Green boxes indicate a significant difference between small and large systems and red boxes failed to reject the null hypothesis. An alpha level of 0.05 is used to determine the significance of p-values.

PFAS Type	chi-squared	df	p-value
All	46.788	1	7.91e-12
PFBA	82.649	1	1.691e-05
PFBS	44.911	1	0.003338
PFHxA	85.577	1	1.063e-05
PFHxS	118.63	1	0.8283
PFOA	17.541	1	0.0009426
PFOS	135.22	1	0.7123

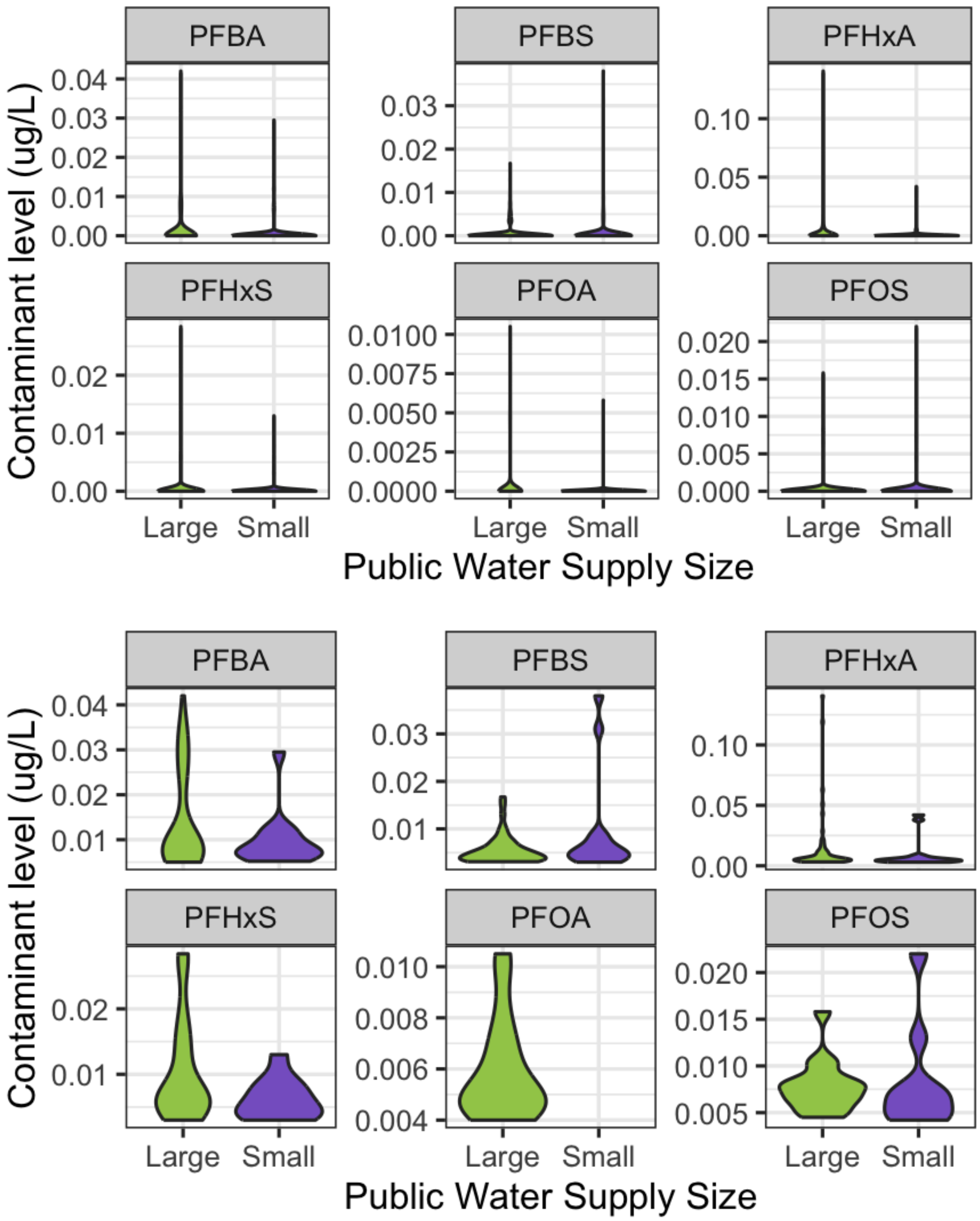

Figure 9. Six PFAS compounds are plotted with the detected level across the public water system (PWS) size. Small is considered <3,000 consumers, large >3,000. The top plot shows contaminant levels, including "<MRL" (Minimum Reporting Level) plotted as 0. The bottom plot filters out "<MRL" showing the distribution of detections.

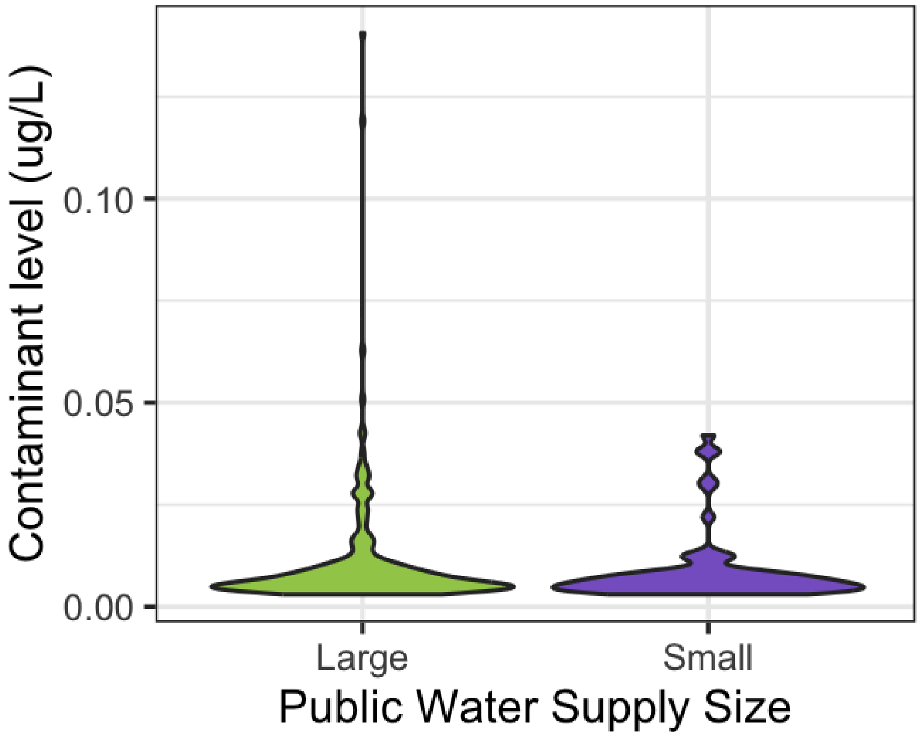

Figure 10. Plots all PFAS contaminant levels separated by public water supply size. Small is considered <3,000 consumers, large >3,000. For this scenario, contaminant levels, including “<MRL” (Minimum Reporting Level) are plotted as 0.

Objective 3

The third objective was to compare PFAS detections between groundwater-supplied and surface water-supplied drinking water facilities. Table 4 displays results from the Wilcoxon rank-sum test, adjusted with Benjamin-Hochberg to control for the false discovery rate. This test yielded six comparisons among combinations of groundwater, surface water, mixed, and groundwater-influenced. The comparison was applied to the six PFAS types and an overall test for all PFAS data. When using all PFAS data, every comparison test is significant (p-values<0.05). The under groundwater influence-surface water comparison (GU-SW) is the only test significant for every PFAS type. GU-GW also had ⅚ of the PFAS types result in significant differences. PFOA had the greatest frequency of insignificant tests ($p>0.05$), with under groundwater influence-surface water being the only significant test (p-value

5.1x10^-5<0.05). Figure __ shows groundwater having the highest reported max contaminant level out of all the facility water types. This is repeatedly observed for every PFAS with the exception of PFBS where under groundwater influence had the highest level.

Table 4. Pairwise comparisons of PFAS concentrations across public water system (PWS) source-water types using Wilcoxon rank-sum tests with continuity correction. P-values were adjusted for multiple comparisons using the Benjamini–Hochberg procedure. Comparisons are among groundwater (GW), groundwater under the influence of surface water (GU), surface water (SW), and mixed sources (MX). Each p-value represents a pairwise comparison between two source-water types, indicating whether PFAS concentrations for a given compound differ between those groups rather than relative to an overall mean across all systems. Cell colors indicate statistical significance: green denotes statistically significant differences (adjusted $p < 0.001$), yellow indicates marginal significance ($0.001 \leq$ adjusted $p < 0.05$), and red indicates no statistically significant difference (adjusted $p \geq 0.05$).

PFAS	Overall	PFBA	PFBS	PFHxA	PFHxS	PFOA	PFOS
GU-GW	2e-16	6.2e-5	7.7e-05	1.6e-05	2.7e-07	0.095	6.2e-05
GU-MX	0.016	0.1	0.018	0.37	0.0024	0.362	0.1
GU-SW	2e-16	5e-8	2.4e-11	6.1e-12	< 2e-16	5.1e-05	5.0e-08
GW-MX	3.4e-08	1.7 e-9	0.954	4.9e-07	0.6046	0.902	1.7e-09
GW-SW	1.5e-08	0.36	0.018	0.02	7.0e-06	0.050	0.36
SW-MX	2e-16	2.1e-15	0.308	2.1e-14	0.0416	0.362	2.1e-15

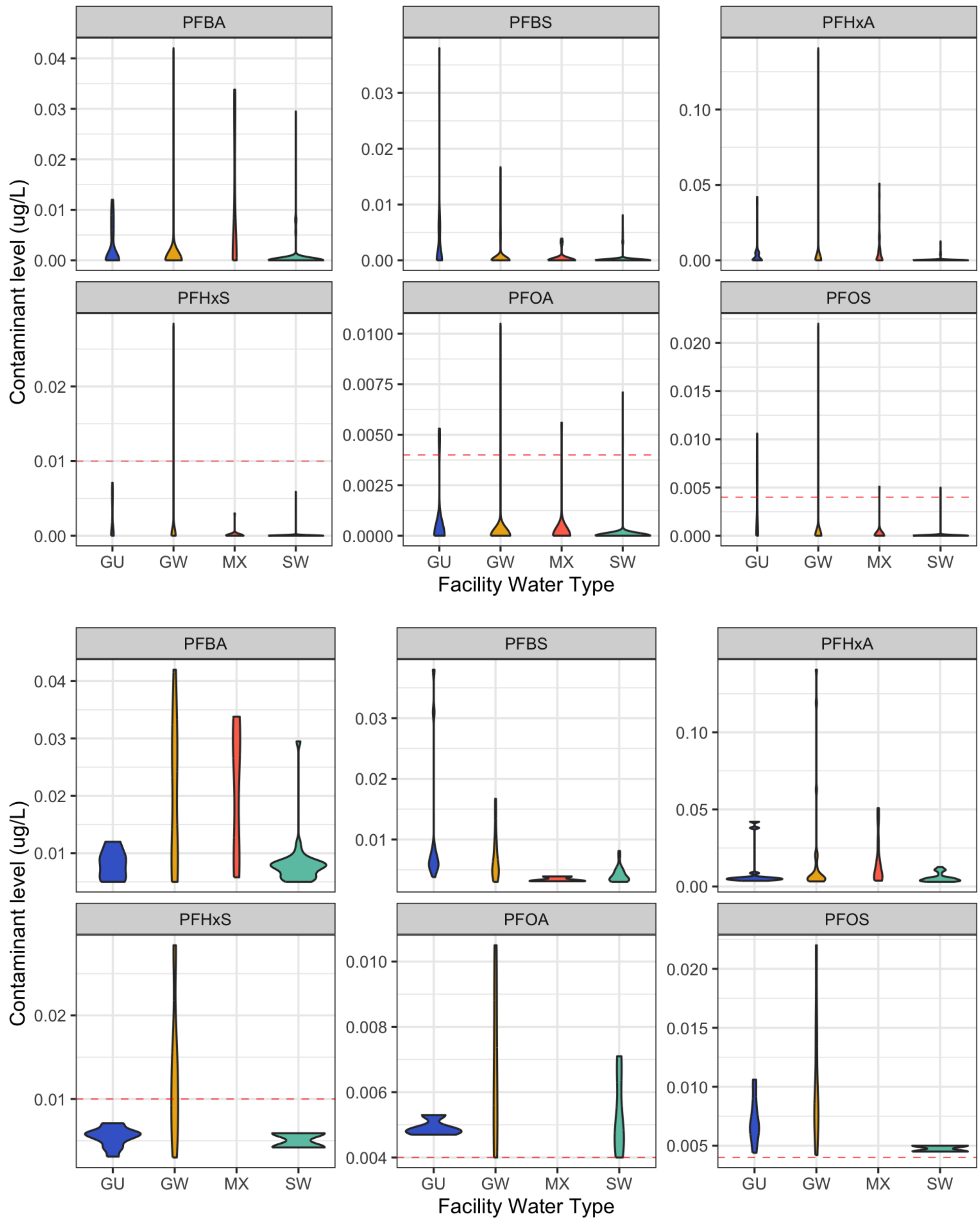

Figure 11. Six PFAS compounds are plotted with the detected contaminant level across different facility water types. These types include groundwater (GW), under groundwater influence (GU), surface water (SW), and mixed (MX). The top plot shows contaminant levels including "<MRL" (Minimum Reporting Level) plotted as 0. The bottom plot filters out "<MRL" showing the distribution of detections. The red lines plotted for PFHxS, PFOA, and PFOS show regulatory limits established for the 2024 PFAS National Primary Drinking Water Regulation.

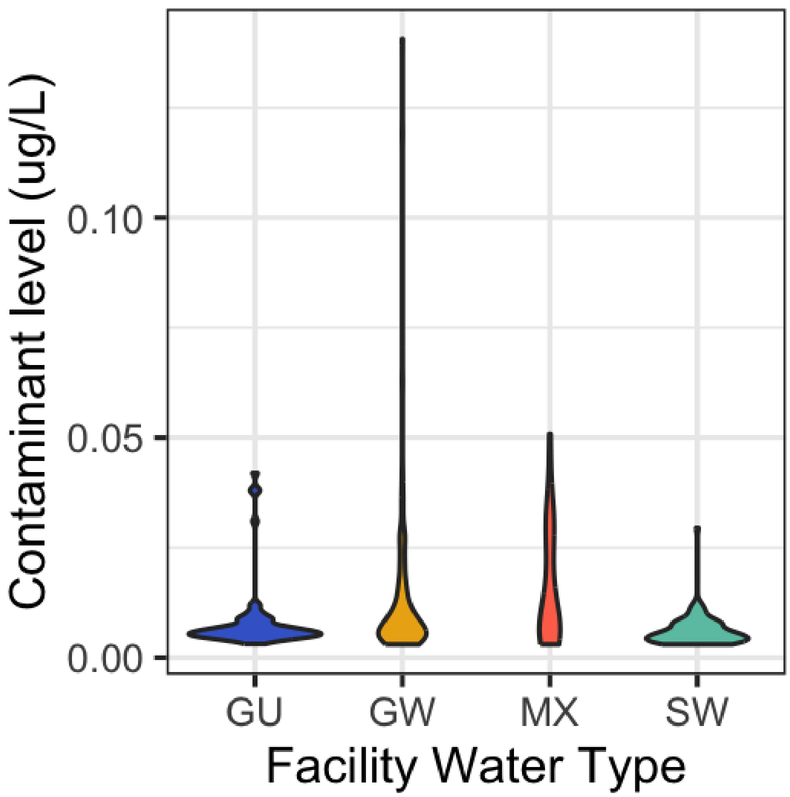

Figure 12. Plots all PFAS contaminant levels separated by facility water type. These categories include groundwater (GW), under groundwater influence (GU), surface water (SW), and mixed (MX). For this scenario, contaminant levels, including "<MRL" (Minimum Reporting Level) are plotted as 0.

Discussion

Spatially plotting PFAS detections revealed patterns regarding where they are typically found. PFAS detections were primarily concentrated between the Denver metropolitan area and southern Colorado Springs. Although detections outside these urban regions were limited, spatial patterns suggest the presence of identifiable local PFAS sources. Patterns across different PFAS types found detections commonly occurring near airports, industry, and ski resorts.

The public water system, Widefield WSD, detected all six PFAS selected for this study. This system is located in an area highly concentrated with military bases— Fort Carson, Cheyenne Mountain Space Force Station, Peterson Space Force Base, the Colorado Springs Airport, and a golf course. Military areas have a history of using firefighting foam containing

PFAS. This association is highlighted in an article linking PFAS detections in U.S. drinking water to military firefighting training areas: an additional military site within a watershed showed a 20% increase in PFHxS, a 10% increase in PFOA, and a 35% increase in PFOS (Hu et al., 2016). Colorado Springs Airport is different from the other airports in Colorado because the Air Force handles firefighting capabilities at the airport (Colorado Department of Transportation, 2021). In accordance with a Department of Defence (DoD) policy, all land-based uses of Aqueous Film Forming Foam (AFFF) containing PFAS substances must be discontinued starting in 2024 (U.S. Department of Defense, 2024). Peterson Space Force Base removed its AFFF in 2017, yet the 2023 data still shows PFAS levels exceeded the minimum risk level. This is likely due to the PFAS compounds, which are characteristically persistent, still leaching into the water supply. It is also possible there are inputs from other companies, such as Microchip Technology Inc. and NuStar Energy Colorado Springs Terminal, which have registered AFFF with the Colorado Department of Public Health and Environment (2026). The legacy of the military using PFAS chemicals has infiltrated from the military bases and into water, soil, groundwater, irrigation ponds, and more. Runoff and leaching from AFFFs are likely why the surrounding areas of Fountain City, Cherokee, Stratmoor Hills, and Fountain City also report contamination.

It is difficult to trace PFAS to a specific source in Parker. The Centennial Airport has likely contributed these chemicals through its use of AFFFs. Additionally, there are numerous golf courses in the area. There is not much academic research connecting PFAS and golf courses, but there have been challenges to the approvals of pesticides containing PFAS for golf courses (Center for Biological Diversity, 2026). It is hard to distinguish pollution from golf courses from contamination with a history of something else. The latter case is observed in Widefield, where a history of Air Force PFAS contamination has accumulated in the ponds used to water the golf course. Future research could investigate this connection to see if PFAS contamination is linked to the chemicals applied to golf courses.

Snake River and Mountain Village are homes to Keystone Resort and Telluride Ski Resort respectively. These ski resorts are likely candidates for why Snake River WD's detected all but one (PFBA) PFAS types. Ski areas are known sources of PFAS contamination due to the use of ski waxes. This is consistent with a German study finding melted snow samples from a ski trail contained PFAS in the µg/L range (Vogel et al., 2024). This pattern does raise questions about why other systems near ski resorts aren't reporting results consistent with these two PWSs. It is possible that the sampling locations could impact results. Samples taken upstream of a ski resort miss the runoff from the mountains, while locations far downstream could be diluted from snowmelt.

Thornton, Todd Creek, and Brighton are all systems located in Adams County, utilizing water from the South Platte watershed. It is difficult to discern a point source of PFAS inputs in this area due to the quantity of potential inputs flowing into the South Platte, not just in Adams County, but also upstream in Denver. Similar to other PWSs discussed in this paper, the history of AFFF use at airports could still be contributing to elevated PFAS levels. This area is home to the primary airports of Colorado, historically the Stapleton airport (1929-1995) and currently Denver International Airport (1995 - present) (Colorado Encyclopedia, n.d.).

Industry inputs are another candidate for contamination in this area, specifically the Suncor Energy Oil Refinery. In 2024, the Suncor refinery received a new state water-pollution permit limiting the quantity of PFAS it can dump into the creek to 70 parts per trillion per day. This permit goes into effect at different times; some limits were enforced by 2025 and some by 2027 (Colorado Department of Public Health and Environment, Water Quality Control Division, 2024). Historically, Suncor has discharged contaminants far exceeding the new regulated level. For example, in June 2023, PFAS discharge from this location was recorded at 2,675 parts per trillion (Phillips, 2024). Interestingly, Thornton, Brighton, and Todd Creek deviated in their detection of PFAS despite using the South Platte as their source water; Thornton and Brighton detected all six PFAS types, while Todd Creek did not detect PFHxA or PFOA. This difference is possibly from differences in facility water type– Thornton uses surface water, Brighton uses groundwater and supplements with water purchased from Thornton, and Todd Creek Village uses alluvial wells. It is hard to certify this relation.

Another potential source of PFAS is Metro Water Recovery, the largest wastewater treatment provider in the Rocky Mountain West (Metro Water Recovery, n.d.). Wastewater treatment plants (WWTPs) are considered conduits because standard treatment processes often fail to remove the PFAS, causing contamination in waterways and biosolids. Monitoring studies across China, Europe, and North America revealed most WWTPs showed low removal for PFAS, with some reporting an increase in the PFAS level after treatment (Lenka et al., 2021). To investigate this WWTP as a possible PFAS source, it would be interesting to sample METROGRO Farm, a dryland farm and pasture located 65 miles east of metropolitan Denver, where biosolids from the WWTP are applied.

This dataset contains samples mostly along the Front Range Urban Corridor and along the main highways. Therefore, there are limitations in the data for individual wells. Based on the high maximum PFAS levels observed at different groundwater sites, these wells could provide important information, especially to the people utilizing them. While there are programs like the Colorado PFAS Testing and Assistance Program (TAP), this is still a pilot program with stipulations possibly preventing people from receiving assistance (Colorado Department of

Public Health and Environment, 2025). Further, well owners may not feel the need to get their water tested because of a history of using the water without issue. While these people may not experience adverse health effects, there could still be PFAS posing a health risk in the groundwater.

As discussed in objective three results, groundwater has the highest maxima of contaminant levels. This statement is supported by PFAS studies in West Virginia, Iowa, and Sweden, which all found the highest contamination levels in groundwater (Banzhaf et al., 2016; Dilparic et al., 2025; McAdoo et al., 2022). Despite these trends, it is hard to make any broad generalizations because everywhere has unique contaminant inputs and geology affecting the movement of PFAS in groundwater. In Colorado, these high levels are likely due to the legacy of firefighting foams used at airports and military bases in Colorado. Due to the persistence of PFAS, these chemicals slowly leach out of these areas rather than flushing out quickly. This longevity creates a difficult task for remediation and can result in accumulation in groundwater.

Objective three indicated a significant difference between small and large systems. There was no overall pattern indicating small or large systems had more frequent or higher detections for contaminants. These differences could be due to the origin of the PFAS. For example, certain chemicals may be more likely to appear in a densely populated residential system than in a more rural agricultural area. Furthermore, larger systems with more consumers could have a higher budget for updating technology to treat certain PFAS chemicals, but these same areas may also have larger inputs of pollutants coming into the system. Further research could investigate why these differences exist for each PFAS in Colorado or could expand the reach to a national dataset to see if any patterns arise.

Conclusion

This study examined how per-and polyfluoroalkyl substance detections in public water systems are spatially distributed across Colorado and how detection levels vary by system size and water source. Most public water systems with PFAS detections are concentrated along Colorado's Front Range Urban Corridor. The strongest contamination patterns appear to be driven more by the sources of PFAS inputs than by differences in system size. Groundwater systems most commonly had the highest reported levels of PFAS.

These findings emphasize the importance of understanding the sources and pathways of PFAS contamination to mitigate future environmental, economic, and public health impacts. While contamination is frequently traced back to industrial or military sources, the financial burden of treatment and infrastructure upgrades often falls on the consumer. This challenge is pronounced for smaller systems or individual well owners who have fewer people to distribute upgrade costs among. While regulators establish standards for PFAS in drinking water, there also needs to be consideration for the creators of these PFAS chemicals. This leaves the question of who will bear the cost of PFAS contamination: the consumer or the polluter?

References

Abunada, Z., Alazaiza, M. Y., & Bashir, M. J. (2020). An overview of per- and polyfluoroalkyl substances (PFAS) in the environment: Source, fate, risk and regulations. *Water, 12*(12), 3590. https://doi.org/10.3390/w12123590

Barton, K. E., Anthamatten, P. J., Adgate, J. L., McKenzie, L. M., Starling, A. P., Berg, K., ... & Richardson, K. (2025). A data-driven approach to identifying PFAS water sampling priorities in Colorado, United States. *Journal of Exposure Science & Environmental Epidemiology, 35*(3), 414–424.

Banzhaf, S., Filipovic, M., Lewis, J., Sparrenbom, C. J., & Barthel, R. (2016). A review of contamination of surface-, ground-, and drinking water in Sweden by perfluoroalkyl and polyfluoroalkyl substances (PFASs). *Ambio, 46*(3), 335–346. https://doi.org/10.1007/s13280-016-0848-8

Center for Biological Diversity. (2026). Lawsuit challenges Trump EPA's latest approval of "forever chemical" pesticide [Press release]. https://biologicaldiversity.org/w/news/press-releases/lawsuit-challenges-trump-epas-latest-approval-of-forever-chemical-pesticide-2026-01-15/

Centers for Disease Control and Prevention. (2025). *PFAS and your health*. Agency for Toxic Substances and Disease Registry. https://www.atsdr.cdc.gov/pfas/about/index.html

Colorado Department of Local Affairs. (2026, January 27). Colorado's state demography office summarizes the U.S. census data released today.

https://cdola.colorado.gov/press-release/colorados-state-demography-office-summarizes-the-us-census-data-released-today

Colorado Department of Local Affairs, Division of Local Government. (n.d.). *Part 1: Planning basics*. https://dlg.colorado.gov/part-1-planning-basics

Colorado Department of Public Health and Environment. (2026, April 17). *Class B firefighting foam containing PFAS: Certificate of registration program* [Data set]. Retrieved April 18, 2026, from https://cdphe.colorado.gov/

Colorado Department of Public Health and Environment. (2025). *PFAS testing and assistance program (PFAS TAP)*. https://cdphe.colorado.gov/pfas/pfas-health/pfas-testing

Colorado Department of Public Health and Environment, Water Quality Control Division. (2024). *Authorization to discharge under the Colorado discharge permit system (Permit No. CO0001147) for Suncor Energy (USA) Inc., Commerce City refinery* [Permit]. https://cdphe.colorado.gov/

Colorado Department of Transportation. (2021). Colorado division of aeronautics completes statewide initiative to minimize environmental impacts of toxic chemicals in firefighting foam at Colorado airports. https://www.codot.gov/news/2021/december/aeronautics-environmental-initiative

Colorado Encyclopedia. (n.d.). *Stapleton International Airport*. https://coloradoencyclopedia.org/article/stapleton-international-airport

Colorado Water Center. (n.d.). *Climate*. Colorado State University. https://waterknowledge.colostate.edu/climate/

Colorado Water Conservation Board. (2023, January 9). *Executive summary: Colorado water plan update*. https://dnrweblink.state.co.us/CWCB/0/edoc/220197/01-09-2023_Executive-Summary.pdf

Dilparic, D., Graesch, M., Hruby, C., & Charbonnet, J. A. (2025). PFAS in Iowa drinking water sources: Chemical and geospatial patterns. *Science of the Total Environment, 977*, 179397. https://doi.org/10.1016/j.scitotenv.2025.179397

Dunnington, D. (2025). *ggspatial: Spatial data framework for ggplot2* (Version 1.1.10) [R package]. https://CRAN.R-project.org/package=ggspatial

Glüge, J., Scheringer, M., Cousins, I. T., DeWitt, J. C., Goldenman, G., Herzke, D., Lohmann, R., Ng, C. A., Trier, X., & Wang, Z. (2020). An overview of the uses of per- and polyfluoroalkyl substances (PFAS). *Environmental Science: Processes & Impacts, 22*(12), 2345–2373. https://doi.org/10.1039/d0em00291g

Hu, X. C., Andrews, D. Q., Lindstrom, A. B., Bruton, T. A., Schaider, L. A., Grandjean, P., Lohmann, R., Carignan, C. C., Blum, A., Balan, S. A., Higgins, C. P., & Sunderland, E. M. (2016). Detection of poly- and perfluoroalkyl substances (PFASs) in U.S. drinking water linked to industrial sites, military fire training areas, and wastewater treatment plants. *Environmental Science & Technology Letters, 3*(10), 344–350. https://doi.org/10.1021/acs.estlett.6b00260

Legislative Council Staff. (2024, March). *A snapshot of Colorado agriculture* (Issue Brief No. 23–286). Colorado General Assembly.

https://content.leg.colorado.gov/sites/default/files/r23-286_agricultural_economy_in_colorado_0.pdf

Lenka, S. P., Kah, M., & Padhye, L. P. (2021). A review of the occurrence, transformation, and removal of poly- and perfluoroalkyl substances (PFAS) in wastewater treatment plants. *Water Research, 199*, 117187. https://doi.org/10.1016/j.watres.2021.117187

McAdoo, M. A., Connock, G. T., & Messinger, T. (2022). *Occurrence of per- and polyfluoroalkyl substances and inorganic analytes in groundwater and surface water used as sources for public water supply in West Virginia* (Scientific Investigations Report No. 2022–5067). U.S. Geological Survey. https://doi.org/10.3133/sir20225067

Metro Water Recovery. (n.d.). *We get your water*. https://www.metrowaterrecovery.com/

Pebesma, E., & Bivand, R. (2023). *Spatial data science: With applications in R*. Chapman and Hall/CRC. https://doi.org/10.1201/9780429459016

Phillips, N. (2024, March 6). Colorado to restrict Suncor's discharge of dangerous "forever chemicals" into Sand Creek. *The Denver Post*. https://www.denverpost.com/2024/03/06/suncor-commerce-city-refinery-water-permit-pfas/

U.S. Department of Defense, Office of the Assistant Secretary of Defense for Energy, Installations, and Environment. (2024). *Guidance for the change out from aqueous film-forming foam to fluorine-free firefighting systems in facilities* [Memorandum].

U.S. Environmental Protection Agency. (2021, December 27). *Fifth unregulated contaminant monitoring rule (UCMR 5)*. https://www.epa.gov/dwucmr/fifth-unregulated-contaminant-monitoring-rule

U.S. Environmental Protection Agency. (n.d.). *Per- and polyfluoroalkyl substances (PFAS) and drinking water*. https://www.epa.gov/sdwa/and-polyfluoroalkyl-substances-pfas

Vogel, C., Roesch, P., Wittwer, P., Sommerfeld, T., Riedel, M., Leube, P., Kalbe, U., Schoknecht, U., & Simon, F.-G. (2024). Per- and polyfluoroalkyl substances (PFAS) in ski waxes and snow from cross-country skiing in Germany: Comparative study of sum parameter and target analysis. *Journal of Hazardous Materials Advances, 16*, 100484. https://doi.org/10.1016/j.hazadv.2024.100484

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York. https://ggplot2.tidyverse.org